

Alusine Bah

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EDUCATION

Islamic University of Technology (IUT)

Bachelor of Science in Electrical & Electronic Engineering

Graduated Oct 2026

Dhaka, Bangladesh

- Coursework: Power Electronics, Energy Conversion, AI & ML, DSP, VLSI, Communication Engineering
- Scholarship: OIC Full Scholarship
- Thesis: Design and performance analysis of an intelligent sensor and locally hosted IoT framework for integrated aquaponics farming (Sub-Saharan Africa case study)

Njala University

Higher National Diploma in Community Health & Clinical Studies

Apr 2021

Bo, Sierra Leone

- Coursework: Clinical Methods, Human Biology, Mental Health, Health Information Management
- Licensed Community Health Officer

EXPERIENCE

Institute of Nuclear Science & Technology

Engineering Intern – Nuclear Energy & Safety

Sept 2025 – Oct 2025

Dhaka, Bangladesh

- Participated in operational procedures within a 3 MW nuclear research reactor facility
- Assisted with reactor diagnostics, radiation monitoring, and safety compliance checks
- Contributed to reactor efficiency assessments and technical safety evaluations

Ghorashal Power Plant

Engineering Intern – Thermal Power Generation

Sept 2025 – Oct 2025

Ghorashal, Bangladesh

- Supported turbine monitoring and maintenance operations in a large-scale thermal power facility
- Assisted engineering teams in evaluating equipment performance and generation efficiency
- Analyzed power distribution processes and identified operational improvement opportunities

Robi Axiata Limited

Engineering Intern – Telecommunications & Network Operations

Sept 2025 – Oct 2025

Dhaka, Bangladesh

- Assisted telecom data centre operations to maintain reliable network performance
- Evaluated network KPIs and identified connectivity improvement opportunities
- Analyzed user data trends to support service planning and demand forecasting

Magbema Islamic Secondary School

Science Teacher

Sept 2017 – 2021

Rokupr, Sierra Leone

- Taught Biology, Chemistry, and Physics to secondary school students
- Prepared students for WASSCE examinations through mentoring and lesson delivery
- Developed curriculum-aligned instructional materials to improve learning outcomes

INTERCEPTS (International College of Professionals & Technocrats)

Teaching Assistant – Department of Nursing

Jan 2020 – 2021

Rokupr, Sierra Leone

- Supported undergraduate nursing instruction in theoretical and practical clinical courses
- Assisted students with research projects, thesis supervision, and academic mentoring
- Contributed to curriculum support and academic quality improvement activities

PROJECTS

Crop Recommendation System

Machine Learning

- ML model recommending suitable crops based on soil nutrients, temperature, humidity, and rainfall

- Built with Python, Scikit-learn, Pandas, and Jupyter; uses real agricultural data

Solar Budget Calculator

Web Tool – JavaScript

- Browser-based tool sizing solar power systems and estimating costs for any country/currency
- Pure HTML, CSS, JavaScript — no dependencies, works offline

IoT Sensor Dashboard

IoT – JavaScript

- Live monitoring dashboard for an agricultural sensor network with 11 simulated sensors
- Includes physics-based cycles, correlations, alerts, and CSV export; designed for Sub-Saharan Africa

Load Flow Solver

Power Systems – Python

- Newton-Raphson and Gauss-Seidel power flow analysis built from first principles
- Per-unit conversion, Y-bus construction, and iterative solvers — verified against Saadat and Stevenson

Power Flow Engineering Calculator

Power Systems – JavaScript

- Browser-based power system analysis tool with Newton-Raphson and Gauss-Seidel load flow methods
- Editable bus and line data, solved voltage profiles, power injections, and contingency studies

ScribeAcademic AI

AI – Web App

- AI document engine converting long education policies, circulars, and printed letters into action checklists
- Accepts PDF, Word, and photos with built-in OCR; designed for schools and universities

Symptom Checker & Disease Encyclopedia

Healthcare – Full-Stack Web App

- Region-aware symptom checker and disease encyclopedia for Africa and other developing regions
- Covers 70 diseases in English and French, with urgency levels and emergency stop for danger signs
- Built with Next.js, FastAPI, Python, SQLite, and TypeScript

IoT Framework for Aquaponics Farming

Engineering Research

- Design and performance analysis of an intelligent sensor and locally hosted IoT framework
- Case study for Sub-Saharan Africa; completed as BSc thesis at IUT, September 2026

Self-Medication Among Youth in Sierra Leone

Public Health Research

- Community-based research assessing self-medication practices among youth
- Applied epidemiological methods, field data collection, and public health analysis

SKILLS

Power Electronics & Energy Systems : Power Electronics, Energy Conversion, Renewable Energy, Power Systems Analysis

Python & Machine Learning : Python, Scikit-learn, Pandas, NumPy, Jupyter, Machine Learning

MATLAB & Data Analysis : MATLAB, Data Analysis, Power BI, Data Visualisation

IoT & Smart Systems : IoT Systems, Embedded Systems, Sensor Integration, Smart Systems

Power BI & Data Visualisation : Power BI, Data Visualisation, Dashboard Design

Community Health & Epidemiology : Community Health, Epidemiology, Public Health, Health Informatics

Circuit Design & Proteus : Circuit Design, Proteus 8, Power Electronics, DSP

Project Management (Google PM) : Project Management, Research, Technical Writing, Leadership

Full-Stack Web (Next.js, FastAPI) : Next.js, FastAPI, JavaScript, TypeScript, HTML, CSS, SQLite

Languages : English (Proficient), Krio (Native), Temne (Native)

CERTIFICATIONS

Google Project Management Professional Certificate
Coursera & Google

Jan 2024
Online

Renewable Energy Technology Fundamentals

July 2023

University of Colorado Boulder & Coursera

Online

Disaster Preparedness

University of Pittsburgh & Coursera

Dec 2023

Online

Industrial Training Certificate

Nuclear, Thermal Power, and Telecom Industrial Placement

2025

Bangladesh

LEADERSHIP & HIGHLIGHTS

International Students Community (ISC), IUT

President

2024 – 2025

Dhaka, Bangladesh

- Led a multicultural student body representing 500+ international students
- Coordinated student welfare, institutional engagement, and community events
- Strengthened collaboration between students and university administration

Sierra Leone Students Community, IUT

President

2023 – 2024

Dhaka, Bangladesh

- Founded and led initiatives supporting Sierra Leonean students in Bangladesh
- Organized academic, cultural, and welfare programs for student integration

International Students Community (ISC), IUT

Deputy Secretary General & Director of Information

2022 – 2024

Dhaka, Bangladesh

- Managed organizational communication and digital information dissemination
- Supported strategic planning, student representation, and event coordination

INTERESTS

Artificial Intelligence, Renewable Energy, Sustainable Technology, Smart Systems, Public Health Innovation, Full-Stack Development, Football

REFEREES

Dr. Mwebesa Umar

Registrar, IUT

- Islamic University of Technology (IUT)
- Email: registrar@iut-dhaka.edu

Engr. Muhammad, SMHSA, MIEB

Assistant Professor, Dept. of EEE

- Islamic University of Technology (IUT)
- Email: muhbdeee@iut-dhaka.edu

Dr. Khondokar Habibul Kabir

Professor, EEE and Chairman of IAEC

- Islamic University of Technology (IUT)
- Email: habib@iut-dhaka.edu